Lecture Scribe CSE400

L15_L17.pdf
PDF

Example Lecture Scribe (Structure Only)
Definitions and Notation
Formal definitions and notation of the lecture.
Assumptions
Every mentioned assumption in the lecture.
Theorem / Results
Explicit statements as they are in the lecture material.
Proof / Derivation
Step-by-step reasoning as shown in the lecture.
Worked Examples
Example problems solved with the intermediate steps as shown.
Conclusion
Key lecture outcomes only.
I am providing you with the material taught in my lecture. You must generate a lecture scribe intended to serve as exam reference material following the structure shown above.
1) Use only the slides given in this zip file and relevant textbook content.
2)Include definitions, notation, assumptions, results, proofs, derivations, and examples only if they are taught in the lecture.
3)Show the steps only till where they are shown in the lecture.
4)Be formal in your academic manner and stick to the flow of the lecture.
5)There should be no explanations, intuition, summaries, new theory or new examples.
6)Do not complete missing steps.
7)Disregard non-formal and pictures.
8)Do not guess or make up anything that you have not been given.

Below is the **Lecture Scribe** generated strictly from the provided slides. Only the formal content explicitly visible in the lecture material has been included, and no additional derivations, explanations, or assumptions have been added.

Lecture Scribe

Course: CSE400 – Fundamentals of Probability in Computing
Lecture: L15–L18
Topic: Joint Random Variables and Joint Distributions
Instructor: Dhaval Patel, PhD
Date: March 10, 2026

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1. Definitions and Notation

Joint Random Variables

In many applications, two or more random variables are considered simultaneously.

Examples of random variables considered jointly:

- X = Vehicle Speed
- Y = Traffic Density
- X = Signal-to-Noise Ratio (SNR)
- Y = Data Throughput
- X = Rainfall
- Y = Temperature
- X = Wind Speed
- Y = Delivery Time
- X, Y = outcomes of rolling two dice

Joint Probability Mass Function (Discrete Case)

For discrete random variables X and Y, the **joint PMF** represents the probability that X and Y simultaneously take specific values.

Notation:

$$p_{X,Y}(x, y) = P(X = x, Y = y)$$

Representation:

- Joint PMF Table Representation

Joint Probability Density Function (Continuous Case)

For continuous random variables X and Y, the joint distribution is described using a **joint probability density function**.

Notation:

$$f_{X,Y}(x, y)$$

Joint Cumulative Distribution Function

The **Joint CDF** of two random variables X and Y is defined as

$$F_{X,Y}(x, y) = P(X \leqslant x, Y \leqslant y)$$

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2. Assumptions

Situations involving multiple random variables may include relationships or correlations between the variables.

Examples provided in the lecture include:

- Vehicle speed and traffic density
- Signal-to-noise ratio and throughput
- Rainfall and temperature
- Wind speed and delivery time

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3. Theorem / Results

Joint PMF Representation

A joint PMF may be represented in **tabular form** where:

- rows correspond to values of X
- columns correspond to values of Y
- table entries represent $P(X = x, Y = y)$.

Joint CDF Definition

$$F_{X,Y}(x, y) = P(X \leqslant x, Y \leqslant y)$$

Joint CDF – Continuous Case

The joint CDF for continuous random variables is obtained from the joint PDF.

Joint CDF – Discrete Case

For discrete random variables, the joint CDF is obtained from the joint PMF.

Properties of the Joint CDF

The lecture introduces properties associated with the joint cumulative distribution function.

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4. Proof / Derivation

No explicit proofs or derivations are formally written in the provided lecture slides.

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5. Worked Examples

Example 1 – Smart Transportation

Random variables:

- X = Vehicle Speed
- Y = Traffic Density

Question:

Can speed be high when the traffic density is also high?

This illustrates correlated random variables.

Example 2 – Wireless Network Performance

Random variables:

- X = Signal-to-Noise Ratio (SNR)
- Y = Data Throughput

Question:

If SNR is high, will throughput always be high?

Example 3 – Weather Prediction

Random variables:

- X = Rainfall
- Y = Temperature

Application: weather forecasting.

Example 4 – Drone Delivery

Random variables:

- X = Wind Speed
- Y = Delivery Time

Application: logistics and UAV systems.

Example 5 – Rolling Two Dice

Two random variables:

- X = outcome of first die

- Y = outcome of second die

Example conditions:

- $X + Y = 7$
- $X > Y$

Joint PMF is required for such events.

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6. Conclusion

The lecture introduces:

- Joint random variables
- Joint PMF for discrete variables
- Joint PDF for continuous variables
- Joint Cumulative Distribution Function
- Applications involving multiple random variables.

If you want, I can also **format this into a clean 1–2 page exam-ready PDF or LaTeX version** (which many students use for CSE400 scribes).